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Robustness properties of a trust region framework for managing approximations in engineering optimization

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ROBUSTNESS PROPERTIES OF A TRUST REGION FRAMEWORK FOR MANAGING APPROXIMATIONS IN ENGINEERING OPTIMIZATION

Work-in-Progress

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Abstract

This paper deals with robustness properties of a framework for managing approximation models of varying fidelity in an engineering optimization process. The model management framework monitors the ability of the lower-fidelity approximation models to predict the behavior of the higher-fidelity models via a trust-region approach. Cheaper, lower-fidelity models are used in the iterative process of generating an optimization step. More computationally expensive, higher-fidelity models are used to update the validity of the resulting design. Both implicit and explicit models can be used. For example, a CFD code can be used as high-fidelity analysis model, while a panel code can be used as a cheaper, lower-fidelity, model. Strong convergence properties of the model management technique are discussed. The approximation management framework will make optimization software more useful for engineering applications.

Introduction

This Work-in-Progress paper deals with the robustness properties of an approach for managing the use of approximate models in optimization proposed in [1]. The approach is also discussed in [4] of these Proceedings.

Briefly, the approximation management framework provides a methodology for making optimization software more useful in engineering applications. Computational models of high fidelity that approximate the reality of a physical process well have reached a certain maturity in many engineering areas. However, the high cost of computation prevents these high-fidelity models from being used in optimization, where reaching a solution may take tens or even hundreds of iterations. On the other hand, cheaper, lower-fidelity models are available, but there is no guarantee that a design that promises improvement when an approximation is used will yield improvement for the higher-fidelity system. The proposed model management framework is designed to address this difficulty.

The model management is easily explained on unconstrained optimization. Let us consider the unconstrained optimization problem

$$\begin{array}{l} \text{minimize } f(x) \\ \text{subject to } x \in \Re^n. \end{array}$$

The proposed method combines the idea of approximation models from engineering design optimization [2] with the model trust region approach from mathematical optimization [5]. The approach can be summarized as follows.

Given a high-fidelity model $f(x)$ of a physical process, and a suite of approximate, lower fidelity models $a_i(x)$, $i = 0, 1, \ldots$ of the same process, use occasionally the the information from the computationally intensive high-fidelity model to correct a lower-fidelity model that is also of lower computational cost. As most optimization iterations are based on the cheaper model, the cost of using expensive function evaluations is minimized.

It should be emphasized that the use of approximations of varying fidelity in a design process is not new. The new contributions of this work are the following:

- In the management of models, we allow both implicit and explicit models or analyses. Function evaluations can be explicit, such as evaluations of linear or quadratic functions. Aerodynamics provides an example of the use of implicit models—a CFD code can serve as a high-fidelity model, while a panel code can serve as a low-fidelity model.

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- A literature search [1] reveals the existing use of models of varying fidelity is done via heuristic procedures in which, as pointed out earlier, one cannot assure the convergence of the process with lower-fidelity models bears a relation to the higher-fidelity process. Moreover, it is not always clear that there is convergence even for the lower-fidelity procedure.

  The proposed model management technique provides a methodology that is substantiated by a rigorous convergence analysis. There is a specific recommendation on when the models should be switched and on the amount of optimization to be done with lower-fidelity models before resorting to higher-fidelity models for guidance.

- This work provides a step toward creating numerical algorithms of practical use in engineering applications.

Various approximation levels can be considered. This paper deals with first order approximations, where the function and derivative evaluations are used in matching the lower-fidelity model to the higher-fidelity model. The use of zeroth order approximations in optimization is addressed in [3].

The Algorithm

Mild requirements are placed on the approximation model a_i used at each optimization iteration:

$$a_i(x_i) = f(x_i) \quad (1)$$

$$\nabla a_i(x_i) = \nabla f(x_i). \quad (2)$$

These are very reasonable requirements: the value of the model a_i and its first derivatives at x_i agree with those of the actual objective f. However, these mild conditions will suffice to prove convergence of the trust region algorithm. They guarantee that sufficiently close to x_i, the approximation a_i is a good model of f. To provide a zeroth order matching is easy, for instance, by scaling. The first order matching, if not available naturally, can be forced in the lower-fidelity model by computing the high-fidelity derivatives at the necessary points. The algorithm is given in Figure 1.

Here "find an approximate solution s_i" means that the solution of each iteration, s_i, can be obtained in any manner suitable to the application, as long as it satisfies a mild decrease condition on the model a_i, namely that it predicts at least a fraction of the decrease in a_i that the steepest descent step inside Δ_i would predict.

Choose $x_0 \in \Re^n$, $\Delta_0 > 0$, $\epsilon > 0$. Choose $\eta_1 \in (0,1)$, $\eta_2 \in (\eta_1, 1)$, $\eta_3 \geq 1$.
For $i = 0, 1, \ldots$ do {
  Compute $f(x_i)$ and $\nabla f(x_i)$.
  Test for convergence ($\| \nabla f(x_i) \| \leq \epsilon$ or other stopping criteria.)
  Choose a_i that satisfies $a_i(x_i) = f(x_i)$ and $\nabla a_i(x_i) = \nabla f(x_i)$.
  Find an approximate solution s_i to:
    minimize $a_i(x_i + s)$
    subject to $\| s \| \leq \Delta_i$
  Set $x_+ = x_i + s_i$ and compare the actual and predicted decrease in f:
  $$R = \frac{f(x_i) - f(x_+)}{f(x_i) - a_i(x_+)}.$$
  If $R < \eta_1$ reject the step,
    set $x_{i+1} = x_i$ and $\Delta_{i+1} = \frac{1}{2}\Delta_i$.
  If $\eta_1 < R < \eta_2$ accept the step
    set $x_{i+1} = x_+$ and $\Delta_{i+1} = \Delta_i$.
  If $R \geq \eta_3$ accept the step
    set $x_{i+1} = x_+$ and $\Delta_{i+1} = 2\Delta_i$.
}

Figure 1: A trust region algorithm using general approximation models.

The algorithm in Figure 2 is guaranteed to produce such predicted decrease. This algorithm is an iterative procedure for solving the problem of minimizing a_i in the algorithm of Figure 1. This is the standard trust-region algorithm for unconstrained minimization. The model a_i is further modeled by the the local quadratic model

$$q(y_j + p) = a_i(y_j) + \nabla a_i(y_j)^T p + p^T H(y_j) p,$$

where y_j is defined in Figure 2 and $H(y_j)$ is the second order information for a_i. $H(y_j)$ can be the true Hessian or its approximation. For instance, it can be zero, and the local model becomes linear.

The conditions (1) and (2) guarantee that sufficiently close to x_i, the approximation a_i is a good model of f. It is then clear how the classical trust region approach provides a mechanism to regulate the use of a_i in an optimization iteration: If the approximation model a_i is not a good predictor of the behavior of f for a long step, we decrease δ and fall back on a region in which a_i is an increasingly good model of f. On the other hand, if a_i is doing a good job of approximating the behavior of f, then we do not decrease the length of the steps we take, and thereby avoid unnecessarily restricting the progress of the optimization. If the prediction is exceptionally good, the size of the step can be increased.

Note that the model need not be fixed across all

Given x_i, $Maxiter > 0$, Δ_i, $\delta_0 < \Delta_i$, set $y_0 = x_i$,
For $j = 0, 1, \ldots, Maxiter$ do {
Take steps to reduce a_i
minimize $q(y_j + p) = \text{model of } a_i$
subject to $\| p \| \leq \delta_j$
$\| y_j + p \| \leq \Delta_i$
to obtain p_j
Compare the actual and predicted decrease in a_i:
$$r = \frac{a_i(x_i) - a_i(y_j + p_j)}{a_i(x_i) - q(y_j + p_j)}$$
Update y_j, δ_j, as x_i and Δ_i of Figure 1.
}
$s_i = p_{Maxiter}$.

Figure 2: A trust region algorithm minimizing a_i approximately.

iterations. In more generality, a_i can be any member of a suite of approximations, as noted earlier. Thus one can change the lower-fidelity model if necessary to reflect the current region of the design space. If a_i is chosen to be a quadratic approximation, then this scheme reduces to the classical model trust region approach.

Furthermore, though only a single level of approximation is described in Figure 1, there is nothing that precludes additional levels of approximation. The framework we propose can accommodate many levels of approximation which vary in fidelity but also, presumably, in computational cost.

Robustness of the Algorithm

In the model management framework, one can say that if an optimization iterate is unsuccessful, the practitioner has an option of selecting a model of increased fidelity or "doing less optimization". While this is a valid general strategy, in order to analyze the framework, a specific strategy—trust regions—is proposed. The trust-region strategy for managing the models places reasonably weak analytical assumption on the procedure. The details of the proof are relegated to [1]. Here we outline the proof of convergence.

In [1], general conditions for global convergence of trust region methods are established. These conditions are listed below together with their resolution for the model management framework. As the conditions are fulfilled, the solution produced by the management algorithm solves the original unconstrained problem.

- **Simple decrease.** Each accepted step in the algorithm of Figure 1 must produce simple decrease in the objective function f, that is, $f(x_{i+1}) < f(x_i)$. In the model management algorithm, simple decrease in f will be obtained by continuity. That is, since a_i and f have the same zeroth and first order information at the current point, then there exists a neighborhood about that point, such that if either a or f attain decrease at another point in that neighborhood, then f or a, respectively, will also attain decrease at that point. The properties of the appropriately chosen trust-region subproblem solution techniques guarantee simple decrease in a. Since the trust-region radius will be decreased in the case of unacceptable steps, after a finite number of steps, the decrease in a_i will imply decrease in f.

- **Sufficient Decrease.** A successful step in a trust-region iteration must predict at least a fraction of the decrease that the steepest descent step would predict on the model of the objective function within the trust region. The first step on a_i satisfies this condition. The rest of the substeps predict even more decrease.

- **Regularity.** It is assumed the high-fidelity objective and the lower-fidelity approximation to be continuous and differentiable functions, which satisfies the regularity requirement.

- **The Zeroth and First Order Information Matching.** It is assumed that the lower-fidelity approximation and the high-fidelity objective have the same function and derivative values at the current point, which is a reasonable assumption to make.

Given the above facts and other standard assumptions of boundedness, convergence to a stationary point of the high-fidelity function can be concluded.

If, in addition, an assumption on the second order information matching at the current point is made, convergence to a local minimum of the high-fidelity objective can be proved.

Current and Future Work

Extension to the constrained case and algorithmic developments are under way. In particular, it is believed that numerical properties of the model management framework can be improved by developing heuristics for testing the progress of the low-fidelity approximation optimization more often. A number of realistic applications have been selected as test cases.

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